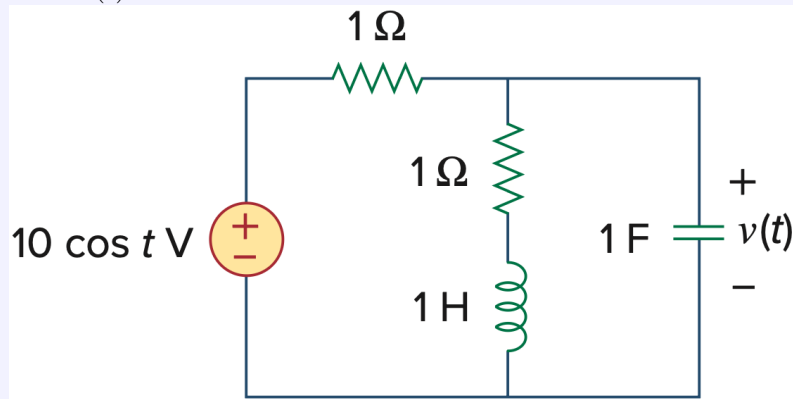


Exercise Sheet #8: AC Networks

Discussion: Dec 8, 2025

Exercise 1: AC Circuit Analysis

Find $v(t)$ in the RLC circuit below.

Solution

We are given the time-domain source

$$v_s(t) = 10 \cos t \text{ V}.$$

In complex (exponential) representation we write this as the real part of a complex exponential:

$$v_s(t) = \operatorname{Re}(\hat{v}_s e^{j\omega t}), \quad \hat{v}_s = 10, \quad \omega = 1 \text{ rad/s},$$

so $v_s(t) = \operatorname{Re}(10e^{jt})$.At $\omega = 1$,

$$\underline{Z}_{R_1} = 1, \quad \underline{Z}_{R_2} = 1, \quad \underline{Z}_L = j\omega L = j, \quad \underline{Z}_C = \frac{1}{j\omega C} = -j.$$

The series $R_2 - L$ branch impedance is

$$\underline{Z}_{RL} = 1 + j.$$

The parallel combination of $\underline{Z}_{RL}$ and $\underline{Z}_C$ has admittance

$$\underline{Y}_p = \frac{1}{\underline{Z}_{RL}} + \frac{1}{\underline{Z}_C} = \frac{1}{1+j} + \frac{1}{-j}.$$

Compute each term explicitly:

$$\frac{1}{1+j} = \frac{1-j}{(1+j)(1-j)} = \frac{1-j}{2} = \frac{1}{2} - \frac{1}{2}j, \quad \frac{1}{-j} = j.$$

Thus

$$\underline{Y}_p = \left(\frac{1}{2} - \frac{1}{2}j\right) + j = \frac{1}{2} + \frac{1}{2}j.$$

Therefore the equivalent impedance of the parallel network is

$$\underline{Z}_p = \frac{1}{\underline{Y}_p} = \frac{1}{\frac{1}{2} + \frac{1}{2}j} = \frac{\frac{1}{2} - \frac{1}{2}j}{(\frac{1}{2})^2 + (\frac{1}{2})^2} = 1 - j.$$

The total series impedance seen by the source is

$$\underline{Z}_{\text{tot}} = \underline{Z}_{R_1} + \underline{Z}_p = 1 + (1 - j) = 2 - j.$$

Now form the complex current $\hat{i}$ (the complex amplitude multiplying e^{jt}) by dividing the complex source amplitude by the complex impedance:

$$\hat{i} = \frac{\hat{v}_s}{\underline{Z}_{\text{tot}}} = \frac{10}{2 - j}.$$

Perform the algebraic division by multiplying numerator and denominator by the conjugate:

$$\hat{i} = 10 \frac{2 + j}{(2 - j)(2 + j)} = 10 \frac{2 + j}{5} = 4 + 2j.$$

Thus the complex current amplitude is $\hat{i} = 4 + 2j$. (Its magnitude is $|\hat{i}| = \sqrt{4^2 + 2^2} \approx 4.472$ and its angle $\arg(\hat{i}) = \arctan(2/4) = 26.57^\circ$, if one prefers polar form.)

The complex voltage across the parallel network (and thus across the capacitor) is

$$\hat{v}_p = \hat{i} \underline{Z}_p = (4 + 2j)(1 - j).$$

Multiply out:

$$\hat{v}_p = 4(1 - j) + 2j(1 - j) = 4 - 4j + 2j - 2j^2 = 4 - 2j + 2 = 6 - 2j.$$

So the complex amplitude of the capacitor voltage is $\hat{v}_p = 6 - 2j$.

We may express this two ways:

(a) Time-domain real part of the exponential:

$$v(t) = \text{Re}(\hat{v}_p e^{jt}) = \text{Re}((6 - 2j)e^{jt}).$$

Expanding $(6 - 2j)e^{jt} = (6 - 2j)(\cos t + j \sin t)$ and taking the real part gives

$$v(t) = 6 \cos t + 2 \sin t.$$

(b) Amplitude–phase form: compute magnitude and angle of $\hat{v}_p$:

$$|\hat{v}_p| = \sqrt{6^2 + (-2)^2} \approx 6.325,$$

$$\arg(\hat{v}_p) = \arctan\left(\frac{-2}{6}\right) \approx -18.43^\circ.$$

Thus

$$\hat{v}_p = 6.325 \angle (-18.43^\circ),$$

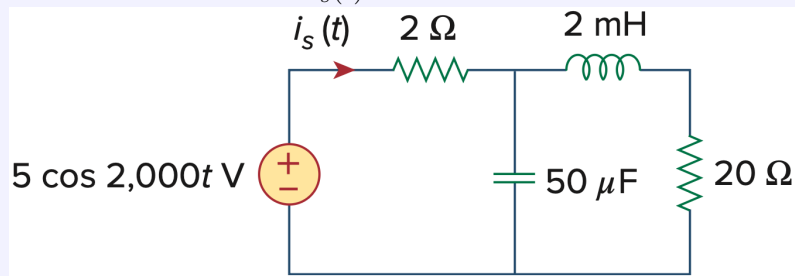
and therefore

$$v(t) = 6.325 \cos(t - 18.43^\circ) \text{ V},$$

which is algebraically equivalent to $v(t) = 6 \cos t + 2 \sin t$.

Exercise 2: ...

Determine the value of $i_s(t)$ in the circuit below.

**Solution**

$$\omega = 2000 \text{ rad/s}$$

$$\underline{Z}_C = \frac{1}{j\omega C} = \frac{-j}{2000 \cdot 50 \times 10^{-6}} \Omega = -j10 \Omega$$

$$\underline{Z}_{RL} = 20 + j\omega L = (20 + j2000 \cdot 2 \times 10^{-3}) \Omega = (20 + j4) \Omega$$

$$\underline{Y}_p = \frac{1}{\underline{Z}_C} + \frac{1}{\underline{Z}_{RL}} = j0.1 + \frac{1}{20 + j4} = j0.1 + \frac{20 - j4}{416} = \left(\frac{20}{416} + j\frac{37.6}{416} \right) \text{ S}$$

$$\underline{Z}_p = \frac{1}{\underline{Y}_p} = \left(\frac{\frac{20}{416} - j\frac{37.6}{416}}{\left(\frac{20}{416}\right)^2 + \left(\frac{37.6}{416}\right)^2} \right) \Omega = (4.58716 - j8.62385) \Omega$$

$$\underline{Z}_{\text{tot}} = \underline{Z}_{R_2} + \underline{Z}_p = 2 \Omega + (4.58716 - j8.62385) \Omega = (6.58716 - j8.62385) \Omega$$

$$\hat{\underline{i}}_s = \frac{\hat{\underline{v}}_s}{\underline{Z}_{\text{tot}}} = \frac{5}{6.58716 - j8.62385} \text{ A} = (0.27714 + j0.36949) \text{ A}$$

$$\hat{i}_s = \sqrt{(0.27714)^2 + (0.36949)^2} \text{ A} = 0.460753 \text{ A}$$

$$\varphi_{i_s} = \tan^{-1}\left(\frac{0.36949}{0.27714}\right) = 52.6263^\circ$$

$$i_s(t) = 460.7 \cos(2000t + 52.63^\circ) \text{ mA}$$

Exercise 3: ...

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Solution

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Exercise 4: ...

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Solution

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Exercise 5: ...

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Solution

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Exercise 6: ...

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Solution

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